

Hardware Components used in the Smart Parking Project:

Component	Quantity	Purpose
ESP32 DevKit V1	1	Main microcontroller for parking management and sensor control
ESP32-CAM (AI Thinker)	1	QR code scanning and camera-based monitoring
IR Sensor	7	1 Entry sensor, 1 Exit sensor, 5 Parking slot occupancy sensors
SG90 Micro Servo Motor	2	Automatic entry and exit gate control
16x2 LCD Display	1	Displays parking and gate status
I2C LCD Module	1	Enables LCD communication with ESP32 using I2C
Red LED	2	Indicates parking full / warning status
Green LED	2	Indicates parking available status
MB102 Breadboard Power Supply Module	1	Provides regulated 5V and 3.3V power
Breadboard	1	Circuit prototyping
Jumper Wires (M-M, M-F, F-F)	as required	Electrical connections
5V 2A SMPS DC Adapter	1	Powers the MB102 and overall circuit
USB Cable (Type-A to Micro-USB)	1	Programming and powering the ESP32 boards
FTDI USB-TTL Programmer	1	Uploading code to the ESP32-CAM
Cardboard Parking Model	1	Prototype Demonstration
Toy Cars	as required	Parking Demonstration